Updated 11/4/26

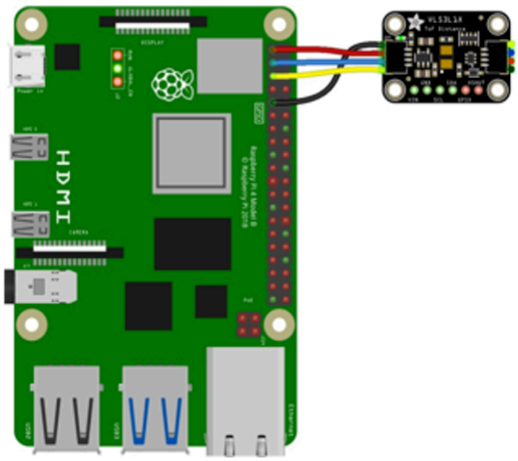
Tunnel Scout How-To Doc

SSH into Raspi

1. Plug in raspi to laptop via USB-C, ensure the correct ends of the fiber optic and ethernet are plugged into the laptop and raspi.
2. Press Win+R, type ncpa.cpl, right click wifi, go to sharing, allow other network users to use this network
    a. Note: wait until the green LED on the raspi stops stuttering and shows a solid light if you want to avoid it possibly hanging on startup or taking a bit longer
3. ssh tunnelscout@raspberrypi.local
    a. Password is tunnelscout1

TOF Collision Sensors

1. SSH onto the raspi
2. source ~/tof/bin/activate
3. Cd CollisionSensors
4. Run python3 tofSensor.py
    a. If you wanted to run it, hit keyboard to exit
5. TOF Sensor Pinout (for one)



Pi 3V to **sensor VIN (red wire)**
Pi GND to **sensor GND (black wire)**
Pi SCL to **sensor SCL (yellow wire)**
Pi SDA to **sensor SDA (blue wire)**

IR Camera

1. SSH onto the raspi
2. Single Photo
    a. rpicam-still -n -o test.jpg creates a test image under test.jpg
        i. In a separate terminal, sftp
        ii. Get test.jpg, then check for it on your device
3. Live Video
    a. run python3 irCam.py
    b. http://192.168.137.80:8000/
    c. If above doesn't work: http://raspberrypi.local:8000/
        i. Ctrl + C to exit

Raspi Pinout

